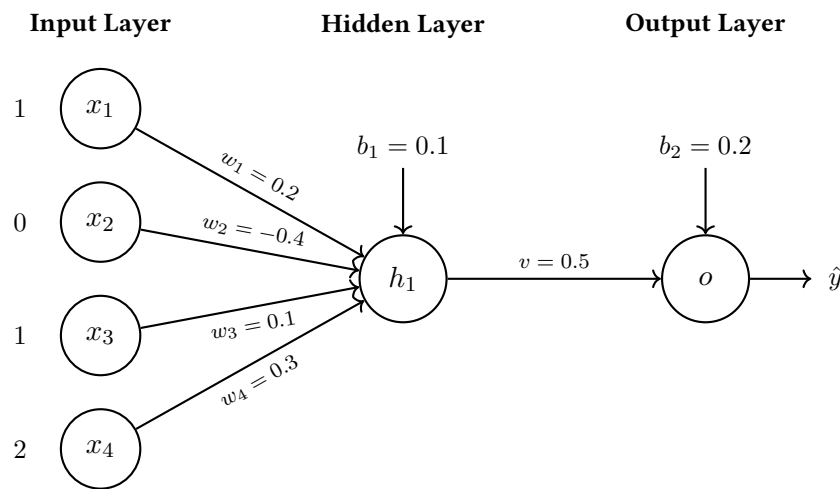


Lecture Notes: Backpropagation in a One-Hidden-Layer Neural Network

Based on the Core Ideas of Chapter 2 of *Neural Networks and Deep Learning*

Dr. Adnan Arefeen

Backpropagation with Four Inputs and One Hidden ReLU Neuron



Derivation of Gradients and Parameter Updates

For the network shown above, we have four inputs, one hidden neuron, and one output neuron.

The target output is:

$$y = 1.$$

The learning rate is:

$$\eta = 0.5.$$

Symbolic Derivation and Computation of All Gradients

In this section, we first derive all gradients symbolically. Then we substitute the numerical values and compute the gradients. Finally, we update the parameters using gradient descent.

Network Equations

The hidden neuron pre-activation is:

$$z_1 = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b_1.$$

The hidden neuron activation is:

$$a_1 = \text{ReLU}(z_1).$$

The output neuron pre-activation is:

$$z_2 = va_1 + b_2.$$

The output activation is:

$$a_2 = \text{ReLU}(z_2).$$

The cost function is:

$$C = \frac{1}{2}(a_2 - y)^2.$$

The ReLU activation function is:

$$\text{ReLU}(z) = \max(0, z).$$

Its derivative is:

$$\text{ReLU}'(z) = \begin{cases} 1, & z > 0, \\ 0, & z < 0. \end{cases}$$

At $z = 0$, ReLU is not differentiable. In practice, we commonly set:

$$\text{ReLU}'(0) = 0.$$

Step 1: Derive All Gradients Symbolically

Output Error Term

The output error term is defined as:

$$\delta_2 = \frac{\partial C}{\partial z_2}.$$

Using the chain rule:

$$\begin{aligned} \delta_2 &= \frac{\partial C}{\partial z_2} \\ &= \frac{\partial C}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2}. \end{aligned}$$

Since

$$C = \frac{1}{2}(a_2 - y)^2,$$

we have:

$$\frac{\partial C}{\partial a_2} = a_2 - y.$$

Also, since

$$a_2 = \text{ReLU}(z_2),$$

we have:

$$\frac{\partial a_2}{\partial z_2} = \text{ReLU}'(z_2).$$

Therefore:

$$\delta_2 = (a_2 - y)\text{ReLU}'(z_2)$$

Gradient with Respect to v

The output neuron input is:

$$z_2 = va_1 + b_2.$$

We want to compute:

$$\frac{\partial C}{\partial v}.$$

Using the chain rule:

$$\begin{aligned}\frac{\partial C}{\partial v} &= \frac{\partial C}{\partial z_2} \cdot \frac{\partial z_2}{\partial v} \\ &= \delta_2 \cdot \frac{\partial}{\partial v}(va_1 + b_2) \\ &= \delta_2 a_1.\end{aligned}$$

Therefore:

$$\frac{\partial C}{\partial v} = \delta_2 a_1$$

Gradient with Respect to b_2

We want to compute:

$$\frac{\partial C}{\partial b_2}.$$

Using the chain rule:

$$\begin{aligned}\frac{\partial C}{\partial b_2} &= \frac{\partial C}{\partial z_2} \cdot \frac{\partial z_2}{\partial b_2} \\ &= \delta_2 \cdot \frac{\partial}{\partial b_2}(va_1 + b_2) \\ &= \delta_2 \cdot 1 \\ &= \delta_2.\end{aligned}$$

Therefore:

$$\frac{\partial C}{\partial b_2} = \delta_2$$

Hidden Error Term

The hidden neuron affects the cost through the following path:

$$z_1 \rightarrow a_1 \rightarrow z_2 \rightarrow a_2 \rightarrow C.$$

The hidden error term is defined as:

$$\delta_1 = \frac{\partial C}{\partial z_1}.$$

Using the chain rule:

$$\begin{aligned}\delta_1 &= \frac{\partial C}{\partial z_1} \\ &= \frac{\partial C}{\partial z_2} \cdot \frac{\partial z_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \\ &= \delta_2 \cdot \frac{\partial}{\partial a_1}(va_1 + b_2) \cdot \frac{\partial}{\partial z_1} \text{ReLU}(z_1) \\ &= \delta_2 \cdot v \cdot \text{ReLU}'(z_1).\end{aligned}$$

Therefore:

$$\boxed{\delta_1 = v\delta_2 \text{ReLU}'(z_1)}$$

Gradients for Input-to-Hidden Parameters

The hidden neuron input is:

$$z_1 = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b_1.$$

The hidden error term is:

$$\delta_1 = \frac{\partial C}{\partial z_1}.$$

Gradient with Respect to w_1

$$\begin{aligned}\frac{\partial C}{\partial w_1} &= \frac{\partial C}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1} \\ &= \delta_1 \cdot \frac{\partial}{\partial w_1}(w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b_1) \\ &= \delta_1 x_1.\end{aligned}$$

Therefore:

$$\boxed{\frac{\partial C}{\partial w_1} = \delta_1 x_1}$$

Gradient with Respect to w_2

$$\begin{aligned}
\frac{\partial C}{\partial w_2} &= \frac{\partial C}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_2} \\
&= \delta_1 \cdot \frac{\partial}{\partial w_2} (w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b_1) \\
&= \delta_1 x_2.
\end{aligned}$$

Therefore:

$$\boxed{\frac{\partial C}{\partial w_2} = \delta_1 x_2}$$

Gradient with Respect to w_3

$$\begin{aligned}
\frac{\partial C}{\partial w_3} &= \frac{\partial C}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_3} \\
&= \delta_1 \cdot \frac{\partial}{\partial w_3} (w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b_1) \\
&= \delta_1 x_3.
\end{aligned}$$

Therefore:

$$\boxed{\frac{\partial C}{\partial w_3} = \delta_1 x_3}$$

Gradient with Respect to w_4

$$\begin{aligned}
\frac{\partial C}{\partial w_4} &= \frac{\partial C}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_4} \\
&= \delta_1 \cdot \frac{\partial}{\partial w_4} (w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b_1) \\
&= \delta_1 x_4.
\end{aligned}$$

Therefore:

$$\boxed{\frac{\partial C}{\partial w_4} = \delta_1 x_4}$$

Gradient with Respect to b_1

$$\begin{aligned}
\frac{\partial C}{\partial b_1} &= \frac{\partial C}{\partial z_1} \cdot \frac{\partial z_1}{\partial b_1} \\
&= \delta_1 \cdot \frac{\partial}{\partial b_1} (w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b_1) \\
&= \delta_1 \cdot 1 \\
&= \delta_1.
\end{aligned}$$

Therefore:

$$\boxed{\frac{\partial C}{\partial b_1} = \delta_1}$$

Step 2: Summary of Symbolic Gradients

$$\delta_2 = (a_2 - y)\text{ReLU}'(z_2)$$

$$\frac{\partial C}{\partial v} = \delta_2 a_1, \quad \frac{\partial C}{\partial b_2} = \delta_2$$

$$\delta_1 = v\delta_2\text{ReLU}'(z_1)$$

$$\frac{\partial C}{\partial w_1} = \delta_1 x_1, \quad \frac{\partial C}{\partial w_2} = \delta_1 x_2$$

$$\frac{\partial C}{\partial w_3} = \delta_1 x_3, \quad \frac{\partial C}{\partial w_4} = \delta_1 x_4, \quad \frac{\partial C}{\partial b_1} = \delta_1$$

Step 3: Compute the Forward Values

The input values are:

$$x_1 = 1, \quad x_2 = 0, \quad x_3 = 1, \quad x_4 = 2.$$

The initial parameters are:

$$w_1 = 0.2, \quad w_2 = -0.4, \quad w_3 = 0.1, \quad w_4 = 0.3,$$

$$b_1 = 0.1, \quad v = 0.5, \quad b_2 = 0.2.$$

The target output is:

$$y = 1.$$

Hidden Neuron

$$\begin{aligned} z_1 &= w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b_1 \\ &= (0.2)(1) + (-0.4)(0) + (0.1)(1) + (0.3)(2) + 0.1 \\ &= 0.2 + 0 + 0.1 + 0.6 + 0.1 \\ &= 1.0. \end{aligned}$$

$$z_1 = 1.0$$

Since $z_1 = 1.0 > 0$,

$$a_1 = \text{ReLU}(z_1) = 1.0.$$

$$a_1 = 1.0$$

Also,

$$\text{ReLU}'(z_1) = 1.$$

Output Neuron

$$\begin{aligned}z_2 &= va_1 + b_2 \\&= (0.5)(1.0) + 0.2 \\&= 0.5 + 0.2 \\&= 0.7.\end{aligned}$$

$$\boxed{z_2 = 0.7}$$

Since $z_2 = 0.7 > 0$,

$$a_2 = \text{ReLU}(z_2) = 0.7.$$

$$\boxed{a_2 = 0.7}$$

Also,

$$\text{ReLU}'(z_2) = 1.$$

Step 4: Compute the Error Terms

Output Error Term

$$\delta_2 = (a_2 - y)\text{ReLU}'(z_2).$$

Substitute the values:

$$\begin{aligned}\delta_2 &= (0.7 - 1)(1) \\&= (-0.3)(1) \\&= -0.3.\end{aligned}$$

$$\boxed{\delta_2 = -0.3}$$

Hidden Error Term

$$\delta_1 = v\delta_2\text{ReLU}'(z_1).$$

Substitute the values:

$$\begin{aligned}\delta_1 &= (0.5)(-0.3)(1) \\&= -0.15.\end{aligned}$$

$$\boxed{\delta_1 = -0.15}$$

Step 5: Compute All Gradients

Output-Layer Gradients

$$\begin{aligned}\frac{\partial C}{\partial v} &= \delta_2 a_1 \\ &= (-0.3)(1.0) \\ &= -0.3.\end{aligned}$$

$$\boxed{\frac{\partial C}{\partial v} = -0.3}$$

$$\begin{aligned}\frac{\partial C}{\partial b_2} &= \delta_2 \\ &= -0.3.\end{aligned}$$

$$\boxed{\frac{\partial C}{\partial b_2} = -0.3}$$

Input-to-Hidden Gradients

$$\begin{aligned}\frac{\partial C}{\partial w_1} &= \delta_1 x_1 \\ &= (-0.15)(1) \\ &= -0.15.\end{aligned}$$

$$\boxed{\frac{\partial C}{\partial w_1} = -0.15}$$

$$\begin{aligned}\frac{\partial C}{\partial w_2} &= \delta_1 x_2 \\ &= (-0.15)(0) \\ &= 0.\end{aligned}$$

$$\boxed{\frac{\partial C}{\partial w_2} = 0}$$

$$\begin{aligned}\frac{\partial C}{\partial w_3} &= \delta_1 x_3 \\ &= (-0.15)(1) \\ &= -0.15.\end{aligned}$$

$$\boxed{\frac{\partial C}{\partial w_3} = -0.15}$$

$$\begin{aligned}\frac{\partial C}{\partial w_4} &= \delta_1 x_4 \\ &= (-0.15)(2) \\ &= -0.30.\end{aligned}$$

$$\frac{\partial C}{\partial w_4} = -0.30$$

$$\begin{aligned}\frac{\partial C}{\partial b_1} &= \delta_1 \\ &= -0.15.\end{aligned}$$

$$\frac{\partial C}{\partial b_1} = -0.15$$

Step 6: Final Gradient List

$$\frac{\partial C}{\partial w_1} = -0.15, \quad \frac{\partial C}{\partial w_2} = 0, \quad \frac{\partial C}{\partial w_3} = -0.15, \quad \frac{\partial C}{\partial w_4} = -0.30$$

$$\frac{\partial C}{\partial b_1} = -0.15, \quad \frac{\partial C}{\partial v} = -0.30, \quad \frac{\partial C}{\partial b_2} = -0.30$$

Step 7: Parameter Updates

The gradient descent update rule is:

$$\theta^{new} = \theta - \eta \frac{\partial C}{\partial \theta}.$$

The learning rate is:

$$\eta = 0.5.$$

Update w_1

$$\begin{aligned}w_1^{new} &= w_1 - \eta \frac{\partial C}{\partial w_1} \\ &= 0.2 - (0.5)(-0.15) \\ &= 0.2 + 0.075 \\ &= 0.275.\end{aligned}$$

$$w_1^{new} = 0.275$$

Update w_2

$$\begin{aligned}w_2^{new} &= w_2 - \eta \frac{\partial C}{\partial w_2} \\ &= -0.4 - (0.5)(0) \\ &= -0.4.\end{aligned}$$

$$w_2^{new} = -0.4$$

Update w_3

$$\begin{aligned}w_3^{new} &= w_3 - \eta \frac{\partial C}{\partial w_3} \\&= 0.1 - (0.5)(-0.15) \\&= 0.1 + 0.075 \\&= 0.175.\end{aligned}$$

$$\boxed{w_3^{new} = 0.175}$$

Update w_4

$$\begin{aligned}w_4^{new} &= w_4 - \eta \frac{\partial C}{\partial w_4} \\&= 0.3 - (0.5)(-0.30) \\&= 0.3 + 0.15 \\&= 0.45.\end{aligned}$$

$$\boxed{w_4^{new} = 0.45}$$

Update b_1

$$\begin{aligned}b_1^{new} &= b_1 - \eta \frac{\partial C}{\partial b_1} \\&= 0.1 - (0.5)(-0.15) \\&= 0.1 + 0.075 \\&= 0.175.\end{aligned}$$

$$\boxed{b_1^{new} = 0.175}$$

Update v

$$\begin{aligned}v^{new} &= v - \eta \frac{\partial C}{\partial v} \\&= 0.5 - (0.5)(-0.30) \\&= 0.5 + 0.15 \\&= 0.65.\end{aligned}$$

$$\boxed{v^{new} = 0.65}$$

Update b_2

$$\begin{aligned}b_2^{new} &= b_2 - \eta \frac{\partial C}{\partial b_2} \\&= 0.2 - (0.5)(-0.30) \\&= 0.2 + 0.15 \\&= 0.35.\end{aligned}$$

$$b_2^{new} = 0.35$$

Final Updated Parameters

After one gradient descent step, the updated parameters are:

$$w_1^{new} = 0.275, \quad w_2^{new} = -0.4, \quad w_3^{new} = 0.175, \quad w_4^{new} = 0.45$$

$$b_1^{new} = 0.175, \quad v^{new} = 0.65, \quad b_2^{new} = 0.35$$